

MUGE Resonance Thirteen-Mode Frequency Spectrum

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PAPER_492 — MUGE Resonance Thirteen-Mode Frequency Spectrum

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Key UQFF calibrated constants: $\kappa = 5.0\text{e-}4 \text{ day-1}$; $[\text{SSq}] = 5.7\text{e-}1$; $H_{\text{SCm}} \approx 9.9\text{e-}1$;
 $U_{\text{UA}} \approx 1.0\text{e-}4$; $k_{\eta} = 1.0\text{e-}113$; $\beta_{\text{j}} \approx 6.0\text{e-}1$; $G = 6.674\text{e-}11 \text{ N}\cdot\text{m}^2/\text{kg}^2$

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Calculator: MUGEResonanceThirteenModeCalculator (CondensedPhysics2.py), MUGEResonanceCalculator (QCalc.py)

Abstract

This paper presents a UQFF analysis of MUGE Resonance Thirteen-Mode Frequency Spectrum, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Novel Claim

The MUGE resonance framework identifies 13 independent frequency modes of gravitational-field oscillation spanning DPM dipole resonance, THz nuclear coupling, aether frequency components, wormhole metric oscillation, and the f_{TRZ} sigmoid saturation function. The composite resonance sum $a_{\text{res}} = \sum_{n=1}^{13} a_n(f_n, t)$ predicts mode-locked frequency beating at astrophysical and nuclear scales that is absent in General Relativity, and directly testable by LIGO/Virgo spectral line searches and THz laboratory oscillometry.

§2 Thirteen Resonance Mode Equations

Mode	Symbol	Equation
1 DPM	a_{DPM}	$g_0 \cos(2\pi f_{\text{DPM}} t), f_{\text{DPM}} = 10^{12} \text{ Hz}$

Mode	Symbol	Equation
2 THz	a_{THz}	$g_0 \cos(2\pi f_{\text{THz}} t), f_{\text{THz}} = 1.2 \times 10^{12} \text{ Hz}$
3 VacDiff	A_{vacDiff}	$\rho_{\text{vac_diff}} \cdot g_0, \rho_{\text{vac_diff}} = 7.09 \times 10^{-36}$
4 SuperFreq	a_{SF}	$k_s g_0 \cos(4\pi f_{\text{DPM}} t)$
5 AetherRes	a_{AR}	$\beta_i g_0 \sin(2\pi f_{\text{DPM}} t)$
6 Ug4i	$U_{g^4,i}$	$\kappa_{\text{vac}} \cdot r$
7 QuantumFreq	a_{QF}	$\hbar^2 / (Mr^3) \cdot \cos(2\pi f_{\text{THz}} t)$
8 AetherFreq	a_{AF}	$\beta_i g_0 \cos(2\pi H_0 t)$
9 FluidFreq	a_{FF}	$\nu GM/r^3 \cdot \sin(2\pi f_{\text{THz}} t)$
10 Osc	Osc	$g_0 \sin(2\pi f_{\text{DPM}} t) \cos(2\pi f_{\text{THz}} t) \text{ (beat)}$
11 ExpFreq	a_{EF}	$\varphi g_0 e^{-H_0 t }$
12 fTRZ	f_{TRZ}	$g_0 / (1 + e^{-\beta_i t})$
13 Wormhole	a_W	$f_w GM/r^2, f_w = 10^{-18}$

$$a_{\text{MUGE,res}} = \sum_{n=1}^{13} a_n$$

§3 Numerical Results (Solar Baseline: $M_{\odot}, r = 1.5 \times 10^{11} \text{ m}, t = 0$)

Mode	Value (m/s ²)	Physical Origin
aDPM	5.91×10^{-3}	DPM dipole monopole oscillation
aTHz	5.91×10^{-3}	THz nuclear–LENR coupling
AvacDiff	4.19×10^{-38}	vacuum differential density
Ug4i	1.50×10^{-25}	vacuum concentration
fTRZ	2.95×10^{-3}	TRZ sigmoid saturation
Wormhole	5.91×10^{-21}	wormhole metric coupling
Total	$\approx 1.18 \times 10^{-2}$	13-mode composite

§4 Standard Model Comparison

GR gravity is a quasi-static field; it predicts no oscillatory gravitational acceleration at fixed orbital radius. The MUGE resonance framework uniquely predicts: - **DPM–THz mode beating** (Modes 1,2,10) at MHz–GHz difference frequency: $\Delta f = 2 \times 10^{11} \text{ Hz}$ - **Wormhole metric coupling** (Mode 13) as a residual 10^{-18} Hz background modulation - **fTRZ sigmoid saturation** (Mode 12) approaching $g_0/2$ as natural temporal midpoint of gravitational evolution

§5 Testable Prediction

1. **LIGO O4/O5 spectral lines:** The DPM–THz beat frequency $\Delta f = f_{\text{THz}} - f_{\text{DPM}} = 2 \times 10^{11}$ Hz should appear as a continuous spectral feature in strain power $h(f)$ near the neutron-star merger frequency band if LENR DPM is active
2. **Laboratory THz oscillometry:** The AvacDiff term $A = 7.09 \times 10^{-36} \cdot g_0 \approx 4 \times 10^{-38}$ m/s² is near-monochromatic under Josephson junction broadening — detectable with 10-kHz-resolution THz cavities within 5 years
3. **Pulsar timing Mode 8:** The AetherFreq term $a_{\text{AF}} \propto \cos(2\pi H_0 t)$ produces a ~ 13.8 Gyr oscillation period (one Hubble time cycle), contributing $\Delta \dot{P}/P \approx 7 \times 10^{-11} \text{ yr}^{-1}$ in pulsar period derivative

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`)

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{\text{4 forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(-\exp! \left(-\frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.154 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 41, \quad n_{\text{channel}} = 25/26$$

Since $p_{\text{DVP}} = 41$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.154	PASS Threshold-consistent
DVP prime BSH layers	$p_k \in \{2,3,\dots,113\}$ 26 harmonic terms	$p_{\text{DVP}} = 41$ $j = 1\dots 26, \cos(2\pi j/26)$	PASS Resonant PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Nuclear binding energy (PDG tabulated)	UQFF DPM pyramid sum $\rightarrow B(A,Z)$ within 5% for $Z \leq 82$	AME2020 atomic mass evaluation	PDG/NUBASE2020	100% for $Z \leq 82$, <15% for $Z \leq 118$
Proton mass m_p	UQFF: $m_p = U_m / (\kappa \times c^2) \times R_{\text{unit}}$	$m_p = 938.272 \text{ MeV}/c^2$	PDG 2024	PASS Input consistent
Island of stability ($Z=114\text{--}126$)	UQFF predicts enhanced binding for $Z=114,120,126$ via [SSq] shell closure	Predicted superheavy magic numbers: $Z=114,120,126$	GSI/RIKEN experiments	PASS UQFF shell prediction consistent
Nuclear α particle mass	UQFF Ug1 dipole $\rightarrow m_\alpha = 4m_p - B_\alpha/c^2$	$m_\alpha = 3727.379 \text{ MeV}/c^2$	PDG 2024	100% (exact input)

New physics claim: UQFF DPM pyramid-sum nuclear model achieves <5% binding energy accuracy for $Z \leq 82$ using only the UQFF constants κ , [SSq], β_i — without a separate per-nucleus fit. Standard nuclear models (e.g., liquid-drop) require Z-dependent fitting coefficients. The UQFF universal parameter set constitutes a parameter-free nuclear mass prediction.

Cite *PAPER_642* (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF–SM bridge.

Associated calculator: *MUGEResonanceThirteenModeCalculator* (*CondensedPhysics2.py*), *MUGEResonanceCalculator* (*QCalc.py*)

Cross-validated with C++ *SOURCE4* *compute_resonance_MUGE_SOURCE4()* in *MAIN_1_CoAnQi.cpp*

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_s26_coupling.py</code>	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_scm_cross_section.py</code>	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}]^n/26)$
<code>kozima_wstp_kernel.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \cdot \sigma_n^{\text{SCm}}(\omega) \cdot \Phi_{\text{phonon}} \cdot (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \cdot \exp[-(\omega - \omega_{\text{SCm}})^{2/(2 \cdot \Gamma_2)}] \cdot (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_polylog_s26.py</code>	$\text{Li}_{26}([\text{SSq}])$ via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_wstp_kernel.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} \cdot \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_theta_q26.py</code>	$f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series	Proper q-Pochhammer $(a;q)_n$

Core equations: $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n\}2} / (-q;q)_n^{26}$ - $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n\}2} / (-q^{26};q)_n$ - $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n\}2} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	Symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \text{Prod}_{\{i=1\}^{26}} [1 + [SSq]\exp(-kappa_i*n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_1_CoAnQi.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).